

just another fast processor—the parallel processing ability is the actual benefit that needs to be leveraged in the long run, and this needs truly parallel applications as well.

2 Virtualisation enters the embedded domain

Venkatesh Kumaran, country manager, Wind River, points out that not just multi-core technology but virtualisation is also gaining momentum. Much has been said and done about server and desktop virtualisation, but now the IT industry has realised that there is a lot of scope for virtualisation in the embedded domain too. The flexibility, parallel processing, enhanced processor utilisation and other benefits of virtualisation have been found to be very useful in mobile phones, tablet PCs and deeply-embedded avionics and automotive applications.

Embedded virtualisation refers to a type-1 hypervisor deployed within an embedded system. Type-1 hypervisor is a special type of operating system that runs directly on the bare hardware. It allows multiple virtual machines or applications to use the underlying hardware platform without bothering about the technical complexity or even knowing that they are sharing a system. Embedded hypervisors are a hybrid of a virtualisation platform and a microkernel, which allows these to support VMs as well as individual applications.

In general, embedded hypervisors implement a different kind of abstraction with different constraints than other platforms. For example, embedded hypervisors must be small and extremely efficient in their use of memory—since memory tends to be one of the key limiters to performance in embedded environments. Within this small footprint, the virtualisation platform also needs to implement a lot of security measures, as embedded systems are often a part of a larger network of devices.

Most importantly, the embedded hypervisor must support scheduling with real-time capabilities. In the case

of handsets, the hypervisor can share the platform with core communication capabilities and third-party applications. Scheduling with real-time characteristics allows the critical functions to coexist with applications that operate on a best-effort basis. Some of the



Wind River's Hypervisor supports Freescale QorIQ P4 platform series

existing embedded virtualisation solutions include PikeOS, OKL4, NOVA and Codezero.

3 Operating system becomes mandatory, software turns more complex

Kumaran points out, "The operating system (OS) is becoming mandatory on every device, and the software code per device has also increased substantially."

Older embedded systems could do away with a formal OS and manage with just small control programs for a limited set of input/output (I/O) and memory services. But, the complexity as well as security and safety concerns of the embedded systems now used in communication systems, consumer electronics, industrial control, automotive, aerospace and defence applications have shifted the focus from lightweight, in-house control systems to

proven, commercial operating systems.

The demand for low latency, low footprint, real-time operating systems (RTOS) that can handle standard functions memory protection, error checking and reporting, and transparent inter-process communications

is increasing. At the same time, the multi-core trend is pushing the capabilities of operating systems further, as there is a need to handle multiple processors, processes, tasks, threads and boards.

Not just the operating systems but the software running on each device has also increased in quantity and complexity. Of course, this poses greater challenges to software engineers

as the development of software for embedded systems has to adhere to a completely different set of rules and priorities—such as greater focus on reducing the footprint and response times. As in the case of operating systems, the multi-core trend is putting a great pressure on software applications too, since they need to be strongly grounded on a 'parallel' thought process.

As in any emerging space, Open Source is playing an important role here too. Open operating systems and tool-chains for embedded systems, such as Android and Eclipse, respectively, are becoming increasingly popular.

4 Increasing security requirements

The security features built into an embedded system are growing into key differentiators today. This is because of the increasing networking capabilities